

The PortPy Fitness Function

This document explains the objective (“fitness”) function that PortPy minimizes when it optimizes a radiation plan. This single number is what our beam-angle-optimization algorithms try to make as small as possible: each candidate set of beam angles is scored by the lowest fitness PortPy can achieve using those angles.

Below I have my written explanation of the scoring function. Following this description I had Claude create the math formulas and provide it’s own explanation if there is trouble reading mine.

The scoring function begins with determining which beams that are going to be active. Each beam has it’s own influence matrix that then gets loaded into the influence matrix once the beam is selected. Each beamlet intensity is going to be stored in vector x . The intensities are gathered in vector x . The influence matrix $A[i,j]$ is the dose deposited within each voxel (i) per 1 unit of intensity in beamlet j .

To create the dose distribution(D) we Multiply matrix A by vector X . $D = A * x$. The dose distribution represents the amount of grey deposited within each voxel. There is a list for each category that contains all of the voxel indices, that are a part of that given category(Lung, PTV, spinal cord, etc.) The amount of gray deposited within each category will then be scored to determine the fitness of the IMRT.

There are 4 sub fitness scores that have a corresponding weight multiplied by their score which will then be summed up to be the final fitness score that we are trying to minimize.

The first is the Ptv underdose function. The weight is 100,000. Which is the strongest weight, which would heavily incentivize for a plan to have the radiation therapy hit the cancerous voxels at the prescribed dose. The formula takes the average of $\text{Max}[0, D - \text{dose } v]^2$. It takes 0, if the prescribed dose is received within the cancerous voxel, however if the dose is under the ptv it will take the difference and square it.

The second is the ptv overdose, It takes $\text{max}[0, \text{overdose of the voxel}]$ and then squares it and averages it. The weight on this function is 10,000 which is the second most important sub function within the fitness function.

The third is OAR sparing, it takes any dose to each healthy voxel and squares it, sum it and divide by the number of voxels within that given organ. We score the dose because the target dose for healthy tissue is 0. Each organ has its own weight that the sub function is multiplied by with the weights ranging from 3-20.

There is a special category of organ sparing which is the rinds. The rinds act as a perimeter around the tumor and have 5 layers. With Rind_0 being the innermost ring, and Rind_4 being the outermost. The rinds contain the same fitness function as the other healthy tissue (averaging the squared gray of every voxel) with the rinds having a smaller weight of 5 for Rind_0 and Rind_1, and a weight of 3 for Rinds 2-4.

It is important to note that voxels can be a part of multiple categories. The tumor recedes within the lung, so tumor voxels will each be within the PTV category as well as the lung category. The rinds will most likely be a part of the lung category.

The fourth measures beam consistency. It takes the difference in intensity between each neighboring horizontal beamlet and squares it. It then sums the squared differences and then multiplies by .6 and divides by the number of columns. The process for the vertical neighbors is the same by squaring the differences of the neighbors then summing them. You multiply by .4 and then divide by the number of rows. You sum these two-sub functions together then multiply by the weight of this function which is 100.

1. The three ingredients

Before the equation, three objects to know:

Symbol	Name	What it is
x	Beamlet intensities	The decision variables. Each beam is a grid of tiny "beamlets" (~2.5 mm pixels); x holds the intensity of every beamlet across all beams (e.g. 2558 values for a 7-beam plan). Constraint: $x \geq 0$ (no negative radiation).
A	Influence matrix	Pre-computed physics. $A[i,j]$ = dose deposited in voxel i by one unit of beamlet j. Shape = (voxels $\times$ beamlets).
$d = A \cdot x$	Dose distribution	The 3D dose delivered to every voxel in the patient — the result of a plan.

Key relationship: **dose = influence matrix $\times$ beamlet intensities** $\rightarrow d = A \cdot x$.

Notation below, for a structure (organ/target) S: A_S = rows of A for S's voxels, so $A_S \cdot x$ = dose to each voxel of S; n_S = number of voxels in S; w_S = that structure's weight; D = target (prescription) dose.

2. The full fitness equation

PortPy minimizes a weighted sum of quadratic penalties:

$$F(x) = \begin{aligned} & (w_{\text{under}} / n_{\text{PTV}}) \cdot \sum [\max(0, D - \text{dose}_v)]^2 && (1) \text{ cover the tumor} \\ & + (w_{\text{over}} / n_{\text{PTV}}) \cdot \sum [\max(0, \text{dose}_v - D)]^2 && (2) \text{ no hot spots} \\ & + \sum_S (w_S / n_S) \cdot \|A_S \cdot x\|^2 && (3) \text{ spare healthy tissue} \\ & + w_{\text{sm}} \cdot (0.6 \cdot \|Q_x \cdot x\|^2 / n_{\text{cols}} + 0.4 \cdot \|Q_y \cdot x\|^2 / n_{\text{rows}}) && (4) \text{ deliverable beams} \end{aligned}$$

where $\text{dose}_v = (A \cdot x)_v$ is the dose to voxel v. The optimizer (MOSEK) finds the x that makes $F(x)$ **smallest**, subject to $x \geq 0$ and a few hard max-dose constraints (Section 6). That minimum value is the fitness score.

Reminder: $\|v\|^2$ just means “square every entry of v and add them up.”

3. Each term, explained

(1) PTV underdose — “cover the tumor” • weight 100,000

$$(w / n_{\text{PTV}}) \cdot \sum [\max(0, D - \text{dose}_v)]^2$$

For every tumor voxel that falls below the prescription D, take the shortfall, square it, average over tumor voxels, and scale by the weight. Voxels at or above D contribute 0. This is the highest-weighted term — covering the tumor is the top priority. (In code this uses a helper variable $dU \geq 0$ with the constraint $A \cdot x \geq D - dU$, so at the optimum dU equals exactly the shortfall.)

(2) PTV overdose — “no hot spots” • weight 10,000

$$(w / n_{\text{PTV}}) \cdot \sum [\max(0, \text{dose}_v - D)]^2$$

The mirror image: penalize tumor voxels above D. Terms (1) and (2) together squeeze every PTV voxel toward exactly the prescription from both sides.

(3) Organ & rind sparing — “keep dose low” • weights 3–20

$$(w_S / n_S) \cdot \|A_S \cdot x\|^2 = (w_S / n_S) \cdot \sum (\text{dose}_v)^2$$

For organs-at-risk (cord, esophagus, heart, lungs) and the rinds (shells expanding outward from the PTV), the target dose is 0, so we penalize any dose, squared. The rinds force the dose to fall off sharply just outside the tumor (conformity).

(4) Smoothness — “make the beam deliverable” • weight 100

$$w_{\text{sm}} \cdot (0.6 \cdot \|Q_x \cdot x\|^2 / n_{\text{cols}} + 0.4 \cdot \|Q_y \cdot x\|^2 / n_{\text{rows}})$$

Each beam's beamlets form a 2D grid (the beam's flat face). Q_x and Q_y compute the difference between neighboring beamlets — left–right (Q_x) and up–down (Q_y). Squaring and summing gives the “jaggedness”; we average by grid size and weight horizontal 0.6 / vertical 0.4 (the collimator leaves slide

horizontally, so that direction matters a bit more for delivery). This keeps fluence maps smooth so the machine can deliver them.

4. Why quadratic (squared)?

Squaring makes large deviations hurt disproportionately more than small ones: a voxel 10 Gy off contributes 100, while ten voxels each 1 Gy off contribute only 10 total. So the optimizer aggressively eliminates big errors and tolerates many tiny ones → smooth, uniform, clinically acceptable plans. It is also convex and differentiable, so MOSEK solves it efficiently. The $1/n$ factors make each term a fair per-voxel average, so a large organ does not dominate just by having more voxels.

5. The weights (lung protocol: Lung_2Gy_30Fx)

The weights encode the clinical priority order. Higher = more important.

Structure	Term type	Weight	Meaning
PTV	quadratic-underdose	100,000	cover the tumor (top priority)
PTV	quadratic-overdose	10,000	no tumor hot spots
ESOPHAGUS	quadratic	20	spare
HEART	quadratic	20	spare
CORD	quadratic	10	spare spinal cord
LUNG_L / LUNG_R / LUNGS_NOT_GTV	quadratic	10	spare lung
RIND_0, RIND_1	quadratic	5	dose fall-off near PTV
RIND_2, RIND_3, RIND_4	quadratic	3	dose fall-off farther out
(fluence)	smoothness-quadratic	100	deliverable beams

Note: the config also lists a linear-overdose term on CORD (weight 100), but the v1.1.4 code does not implement linear-overdose — it is silently skipped. So the cord is protected only by its quadratic (weight 10) term plus the rind max-dose constraints, not the “100” the config suggests.

6. Hard constraints (enforced, not penalized)

Separate from the penalties, the optimizer enforces max-dose limits on the rinds as hard constraints, forcing the dose gradient to drop with distance from the tumor:

Structure	Max dose limit
RIND_0 (PTV + 5 mm shell)	$1.10 \times \text{prescription}$
RIND_1	$1.05 \times \text{prescription}$
RIND_2	$0.90 \times \text{prescription}$

Structure	Max dose limit
RIND_3	0.85 × prescription
RIND_4	0.75 × prescription

7. Where this lives in the code

What	Location
Objective terms assembled	portpy/photon/optimization.py : 111–143 (create_cvxpy_problem)
Objective minimized	optimization.py : 405 → cp.Problem(cp.Minimize(cp.sum(self.obj)), ...)
Fitness value returned	optimization.py : 425 → self.obj_value = problem.value
Weights / structures / constraints	config_files/optimization_params/optimization_params_Lung_2Gy_30Fx.json
Config loader	data_explorer.py : 358 (load_config_opt_params)

Only four term types are implemented: quadratic-overdose, quadratic-underdose, quadratic, smoothness-quadratic.